

Safety Data Sheet
acc. to OSHA HCS

Printing date 02/06/2026

Reviewed on 02/06/2026

1 Identification

· **Product identifier**

· **Trade name:** Kentan® DF

· **Article number:** US24

· **CAS Number:**

EPA Registration No.: 10163-395

Active Ingredient: Copper Hydroxide(61.3 %), CAS: 20427-59-2

· **Application of the substance / the mixture** Fungicide/Bactericide For Agricultural Use

· **Details of the supplier of the safety data sheet**

· **Manufacturer/Supplier:**

Gowan Company, LLC.

P.O. Box 5569

Yuma, Arizona 85366-5569

(928) 783-8844

· **Information department:** sds@gowanco.com

· **Emergency telephone number:**

Chemtrec® Emergency Telephone 24 - Hours: (Spills, leak or fire) Inside U.S. & Canada: (800) 424-9300

Outside the U.S. & Canada: +011 (703) 527-3887

For medical emergency (ProPharma Group®): (888) 478-0798

2 Hazard(s) identification

· **Classification of the substance or mixture**



GHS05 Corrosion

Eye Damage 1

H318 Causes serious eye damage.



GHS07

Acute Toxicity - Oral 4

H302 Harmful if swallowed.

Acute Toxicity - Inhalation 4 H332 Harmful if inhaled.

· **Label elements**

· **GHS label elements**

The product is classified and labeled according to the Globally Harmonized System (GHS).

· **Hazard pictograms**



GHS05



GHS07

· **Signal word** Danger

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- **Hazard-determining components of labeling:**
Copper hydroxide
- **Hazard statements**
H302+H332 Harmful if swallowed or if inhaled.
H318 Causes serious eye damage.
- **Precautionary statements**
P261 Avoid breathing dust/fume/gas/mist/vapors/spray
- **Hazard description:**
Causes irreversible eye damage. Harmful if swallowed. Harmful if absorbed through skin. Harmful if inhaled. Do not get in eyes or on clothing. Avoid contact with skin. Avoid breathing dust.
- **Other hazards**
- **Results of PBT and vPvB assessment**
 - **PBT:** Not applicable in US.
 - **vPvB:** Not applicable in US.

3 Composition/information on ingredients

- **Chemical characterization: Mixtures**
- **Description:** Mixture: consisting of the following components.

· Dangerous components:		
CAS: 20427-59-2	Copper hydroxide  Acute Toxicity - Inhalation 2, H330;  Aquatic Acute 1, H400 (M=10);  Acute Toxicity - Oral 4, H302; Acute Toxicity - Dermal 4, H312	61.3%

- **Additional information:** Metallic Copper (Cu²⁺) Equivalent is 40% by weight

4 First-aid measures

- **Description of first aid measures**
- **General information:**
Have the product container or label with you when calling a poison control center or doctor, or going for treatment.
You may also contact 1-888-478-0798 for emergency medical treatment information.
Si usted no entiende la etiqueta, busque a alguien para que se la explique a usted en detalle.
(If you do not understand the label find someone to explain it to you in detail.)
- **After inhalation:**
 - Move person to fresh air.
 - If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth, if possible.
 - Call poison control center or doctor for further treatment advice.
- **After skin contact:**
 - Take off contaminated clothing.
 - Rinse skin immediately with plenty of water for 15-20 minutes.
 - Call a poison control center or doctor for treatment advice.
- **After eye contact:**
 - Hold eye open and rinse slowly and gently with water for 15-20 minutes.
 - Remove contact lenses, if present, after first 5 minutes, then continue rinsing eyes.
 - Call a poison control center or doctor for treatment advice.
- **After swallowing:**
 - Call a poison control center or doctor immediately for treatment advice.
 - Have person sip a glass of water if able to swallow.
 - Do not induce vomiting unless told to do so by a poison control center or doctor.
 - Do not give anything by mouth to an unconscious person.

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- **Information for doctor:** Probable mucosal damage may contraindicate use of gastric lavage.
- **Most important symptoms and effects, both acute and delayed** Unknown
- **Indication of any immediate medical attention and special treatment needed** None

5 Fire-fighting measures

- **Extinguishing media**
 - **Suitable extinguishing agents:** CO₂, extinguishing powder or water spray. Fight larger fires with water spray.
- **Special hazards arising from the substance or mixture**
 - Do not inhale explosion and combustion gases.
 - Burning produces heavy smoke.
- **Advice for firefighters**
 - Firefighters and others that may be exposed to vapors, mist, dusts, or products of combustion should wear full protective clothing and self-contained breathing apparatus.
 - Equipment should be thoroughly cleaned after use
 - **Protective equipment:**
 - Mouth respiratory protective device.
 - Wear self-contained respiratory protective device.

6 Accidental release measures

- **Personal precautions, protective equipment and emergency procedures**
 - Wear protective equipment. Keep unprotected persons away.
 - Keep unnecessary people away and isolate hazard area. Wear appropriate personal protective equipment described in Section 8.
- **Environmental precautions:**
 - This pesticide is toxic to fish and aquatic invertebrates and may contaminate water through runoff. This product has a potential for runoff for several months or more after application. Poorly draining soils and soils with shallow water tables are more prone to produce runoff that contains this product.
 - Drift and runoff may be hazardous to aquatic organisms in water adjacent to treated areas.
 - Do not apply directly to water, to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not contaminate water when disposing of equipment washwater or rinsate.
- **Methods and material for containment and cleaning up:**
 - Use neutralizing agent. Dispose of contaminated material as waste according to item 13.
 - Ensure adequate ventilation. Sweep up spill, scoop or vacuum and remove.
 - Flush residual spill area with water.
 - Contain large spills with dikes and transfer the material to appropriate containers for reclamation or disposal.
 - Absorb remaining material or small spills with an inert material and then place in a chemical waste container.
 - Flush residual spill area with water.
 - Refer to section 13 for disposal information and section 15 for reportable quantity information.
 - Wash with plenty of water.
- **Reference to other sections**
 - See Section 7 for information on safe handling.
 - See Section 8 for information on personal protection equipment.
 - See Section 13 for disposal information.
- **Protective Action Criteria for Chemicals**

· PAC-1:

All components have the value 1.5 mg/m³.

· PAC-2:

All components have the value 16 mg/m³.

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· **PAC-3:**All components have the value 97 mg/m³.

7 Handling and storage

· **Handling:**· **Precautions for safe handling**

- Wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet.
- Remove PPE immediately after handling this product.
- Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing.
- Remove clothing immediately if pesticide gets inside. Then wash thoroughly and change into clean clothing.
- **Information about protection against explosions and fires:** No special measures required.

· **Conditions for safe storage, including any incompatibilities** Keep away from food, drink, and feed.· **Storage:**· **Requirements to be met by storerooms and receptacles:**

Store under well-vented, cool and dry storage conditions. Do not store under moist conditions.

· **Information about storage in one common storage facility:** Store away from foodstuffs.· **Further information about storage conditions:** Keep receptacle tightly sealed.· **Specific end use(s)** No further relevant information available.

8 Exposure controls/personal protection

· **Additional information about design of technical systems:** No further data; see section 7.· **Control parameters**· **Components with limit values that require monitoring at the workplace:**

The product does not contain any relevant quantities of materials with critical values that have to be monitored at the workplace.

· **Regulatory information TLV:** Guide to Occupational Exposure Values (TLV)· **Additional information:** The lists that were valid during the creation were used as basis.· **Exposure controls**· **Personal protective equipment:**· **General protective and hygienic measures:**

Wash hands before breaks and at the end of work.

Keep away from foodstuffs, beverages and feed.

Immediately remove all soiled and contaminated clothing.

Avoid contact with the eyes and skin.

Users should wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet. Users should remove clothing/PPE immediately if pesticide gets inside. Then wash thoroughly and put on clean clothing. Users should remove PPE immediately after handling this product. As soon as possible, wash thoroughly and change into clean clothing. Wash the outside of gloves before removing.

· **Breathing equipment:**

In case of brief exposure or low pollution use respiratory filter device. In case of intensive or longer exposure use respiratory protective device that is independent of circulating air.

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· Protection of hands:



Protective gloves

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation. Some materials that are chemical-resistant to this product are listed below. If you want more options, follow the instructions for category A on an EPA chemical-resistance category selection chart.

· Material of gloves

chemical-resistant gloves such as Natural Rubber.

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer. As the product is a preparation of several substances, the resistance of the glove material can not be calculated in advance and has therefore to be checked prior to the application.

· Penetration time of glove material

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

· Eye protection:



Tightly sealed goggles

· Body protection:

Handlers must wear:

- long-sleeved shirt and long pants
- shoes plus socks
- goggles or faceshield
- chemical-resistant gloves such as Natural Rubber.

9 Physical and chemical properties

· Information on basic physical and chemical properties

· General Information

· Appearance:

- Form: Granulate
- Color: Light blue-green
- Odor: Odorless
- Odor threshold: Not determined

· pH-value: 8,8 - 9,0

· Change in condition

- Melting point/Melting range: Undetermined
- Boiling point/Boiling range: Undetermined

· Flash point: Not applicable

· Flammability: Not determined

· Decomposition temperature: Not determined

· Ignition temperature: Product is not self-igniting.

· Danger of explosion: Product does not present an explosion hazard.

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· **Explosion limits:**

- **Lower:** Not determined
- **Upper:** Not determined

· **Vapor pressure:** Not applicable· **Density:** 1.02 g/cm³ (8.5119 lbs/gal)

- **Relative density** Not determined
- **Vapor density** Not applicable
- **Evaporation rate** Not applicable.

· **Solubility in / Miscibility with**

- **Water:** Insoluble.

· **Partition coefficient (n-octanol/water):** Not determined· **Viscosity:**

- **Dynamic:** Not applicable
- **Kinematic:** Not applicable

· **Other information** No further relevant information available.

10 Stability and reactivity

· **Reactivity** No further relevant information available.· **Chemical stability**

This product should be stable for at least two years under normal conditions of warehouse storage. Store under well-vented, cool and dry storage conditions away from foodstuffs, reducing agents and acids.. Do not store under moist conditions.

- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.

- **Possibility of hazardous reactions** No dangerous reactions known.

- **Conditions to avoid** Store under well-vented, cool and dry storage conditions, Avoid moisture conditions.

- **Incompatible materials:** No further relevant information available.

- **Hazardous decomposition products:** No dangerous decomposition products known.

11 Toxicological information

· **Information on toxicological effects**· **Acute toxicity:**· **LD/LC50 values that are relevant for classification:**

Oral	LD50	944.6 mg/kg (rat)
Dermal	LD50	>2,000 mg/kg (rat)
Inhalation	LC50	>1.28 mg/l (rat; 4hr)

CAS: 20427-59-2 Copper hydroxide

Oral	LD50	500 mg/kg (ATE)
Inhalative	LC50/4 h	0.47 mg/l (ATE)

0.47 mg/l (Acute toxic estimated)

· **Primary irritant effect:**

- **on the skin:** No irritant effect.

- **on the eye:**

Moderately Irritating

Strong irritant with the danger of severe eye injury.

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- **Sensitization:** No sensitizing effects known.
- **Additional toxicological information:**
The product shows the following dangers according to internally approved calculation methods for preparations:
Harmful
Irritant

- **Carcinogenic categories**

- **IARC (International Agency for Research on Cancer)**

None of the ingredients are listed.

- **NTP (National Toxicology Program)**

None of the ingredients are listed.

- **OSHA-Ca (Occupational Safety & Health Administration)**

None of the ingredients are listed.

12 Ecological information

- **Toxicity**

This pesticide is toxic to fish and aquatic invertebrates and may contaminate water through runoff. This product has a potential for runoff for several months or more after application. Poorly draining soils and soils with shallow water tables are more prone to produce runoff that contains this product. Drift and runoff may be hazardous to aquatic organisms in water adjacent to treated areas.

Do not apply directly to water, to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not contaminate water when disposing of equipment washwater or rinsate.

- **Aquatic toxicity:** No further relevant information available.

- **Persistence and degradability**

Data referred to copper hydroxide tech. grade:

Stable to hydrolysis and not expected to be degraded by photolysis in water.

Not readily biodegradable.

BOD: not applicable

COD: not available

- **Behavior in environmental systems:**

- **Bioaccumulative potential** No further relevant information available.

- **Mobility in soil** No further relevant information available.

- **Additional ecological information:**

- **General notes:**

Water hazard class 2 (Self-assessment): hazardous for water

Do not allow product to reach ground water, water course or sewage system.

Must not reach bodies of water or drainage ditch undiluted or unneutralized.

Danger to drinking water if even small quantities leak into the ground.

- **Results of PBT and vPvB assessment**

- **PBT:** Not applicable

- **vPvB:** Not applicable

- **Other adverse effects** None

13 Disposal considerations

- **Waste treatment methods**

- **Recommendation:**

Do not contaminate water, food, or feed by storage and disposal.

Wastes resulting from the use of this product must be disposed of on site or at an approved waste disposal facility.

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· **Waste disposal key:** This is a nonrefillable, non-rigid container (bag). Do not reuse or refill this container.

· **Uncleaned packagings:**

· **Recommendation:**

Nonrefillable Paper and Plastic Bags

Nonrefillable container. Do not reuse or refill this container. Completely empty bag into application equipment, then offer for recycling if available or dispose of empty bag in a sanitary landfill or by incineration.

Nonrefillable Fiber Drums with Liners

Nonrefillable container. Do not reuse or refill this container. Completely empty the liner by shaking and tapping sides and bottom to loosen clinging particles. Empty residue into application equipment, then offer for recycling if available or return to the Seller, or dispose of in a sanitary landfill or by incineration or if allowed by state and local authorities, by burning. If burned, stay out of smoke. Fiber drum is to be disposed of in the same manner required for its liner.

14 Transport information

· **UN-Number**

· **DOT**

Void

· **ADR, IMDG, IATA**

UN3077

· **UN proper shipping name**

· **DOT**

Void

· **ADR**

3077 ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Copper hydroxide)

· **ADN**

Not regulated

· **IMDG**

ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Copper hydroxide), MARINE POLLUTANT

· **IATA**

ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Copper hydroxide)

· **Transport hazard class(es)**

· **DOT**

· **Class**

Void

· **ADR, IATA**



· **Class**

9 Miscellaneous dangerous substances and articles

· **Label**

9

· **IMDG**



· **Class**

9 Miscellaneous dangerous substances and articles

· **Label**

9

· **Packing group**

· **DOT**

Void

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· ADR, IMDG, IATA	III
· Environmental hazards:	
· Marine pollutant:	No
· Special marking (ADR):	Symbol (fish and tree)
· Special marking (IATA):	Symbol (fish and tree)
· Special precautions for user	Warning: Miscellaneous dangerous substances and articles
· Hazard identification number (Kemler code):	90
· EMS Number:	F-A, S-F
· Stowage Category	A
· Stowage Code	SW23 When transported in BK3 bulk container, see 7.6.2.12 and 7.7.3.9.
· Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	Not applicable
· Transport/Additional information:	
· ADR	
· Excepted quantities (EQ)	Code: E1 Maximum net quantity per inner packaging: 30 g Maximum net quantity per outer packaging: 1000 g
· IMDG	
· Limited quantities (LQ)	5 kg
· Excepted quantities (EQ)	Code: E1 Maximum net quantity per inner packaging: 30 g Maximum net quantity per outer packaging: 1000 g
· UN "Model Regulation":	US DOT: Not Regulated
	Others: UN 3077 ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COPPER HYDROXIDE), 9, III

15 Regulatory information

· Safety, health and environmental regulations/legislation specific for the substance or mixture

EPA /FIFRA Information:

This chemical is a pesticide product registered by the Environmental Protection Agency (EPA) and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals.

The following is the hazard information as required on the pesticide label.

· Marketing authorization number:

· SARA Title III

· Section 355 (extremely hazardous substances):

None of the ingredients are listed.

· Section 313 (Specific toxic chemical listings):

All ingredients are listed.

· TSCA (Toxic Substances Control Act):

All components have the value ACTIVE.

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· **Hazardous Air Pollutants**

None of the ingredients are listed.

· **Proposition 65**· **Chemicals known to cause cancer:**

None of the ingredients are listed.

· **Chemicals known to cause reproductive toxicity for females:**

None of the ingredients are listed.

· **Chemicals known to cause reproductive toxicity for males:**

None of the ingredients are listed.

· **Chemicals known to cause developmental toxicity:**

None of the ingredients are listed.

· **Carcinogenicity categories**· **EPA (Environmental Protection Agency)**

None of the ingredients are listed.

· **TLV (Threshold Limit Value)**

None of the ingredients are listed.

· **NIOSH-Ca (National Institute for Occupational Safety and Health)**

None of the ingredients are listed.

· **GHS label elements**

The product is classified and labeled according to the Globally Harmonized System (GHS).

· **Hazard pictograms**

Not applicable

· **Signal word**

(US EPA) DANGER

· **Hazard-determining components of labeling:**

Copper hydroxide

· **Hazard statements**

DANGER - PELIGRO

Causes irreversible eye damage. Harmful if swallowed. Harmful if absorbed through skin. Harmful if inhaled. Do not get in eyes or on clothing. Avoid contact with skin. Avoid breathing dust.

· **Precautionary statements**

P261 Avoid breathing dust/fume/gas/mist/vapors/spray

· **Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

16 Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

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· **NFPA ratings (Scale 0-4)**

Health = 2

Fire = 0

Reactivity = 0

· **Department issuing SDS:** Supply Chain· **Contact:** sds@gowanco.com· **Date of preparation / last revision** 02/06/2026 / 18.0· **Abbreviations and acronyms:**

ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

DOT: US Department of Transportation

IATA: International Air Transport Association

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

·

NIOSH: National Institute for Occupational Safety

OSHA: Occupational Safety & Health

TLV: Threshold Limit Value

PEL: Permissible Exposure Limit

REL: Recommended Exposure Limit

Acute Toxicity - Oral 4: Acute toxicity – Category 4

Acute Toxicity - Inhalation 2: Acute toxicity – Category 2

Eye Damage 1: Serious eye damage/eye irritation – Category 1

Aquatic Acute 1: Hazardous to the aquatic environment - acute aquatic hazard – Category 1

· **Sources** Kentan® DF is a registered trademark of Gowan Company.